

feature cutting · lightweight prediction · metric-aware calibration

Data

● _____

● _____

● _____

 facial cues: gaze · AU

a_x

a_y

a_z

36-d

12 raw $\rightarrow$ 3 acceleration axes

gait stats: band energy · autocorr peak
· step interval · axis coupling

Big Five (5-d)

5-d

Personality Trait	Score (approx.)
O	4.5
C	3.5
E	2.5
A	3.5
N	3.0

Diagram illustrating the selection of features for the AU intensity and AU presence tasks. A funnel icon indicates the selection process. The 'keep' features are AU intensity, AU presence, and gaze. The 'discard' features are landmarks, head pose, and shape. Below, a bar chart shows AU intensity and a sequence of colored squares shows AU presence, with a gaze (2D) icon below.


```

graph LR
    A["3-axis gait + Big Five 41-d"] --> B["Standard Scaler"]
    B --> C["Ordinal Ridge"]
    C --> D["orig"]
  
```

E1 E2 E3 three task events

auvel
all-event mean · std · Δ frame

e3only
terminal event-3 state

fatigue
cross-event drift $\mu^{(e)}$ $\rightarrow$ drift $f_{fatigue}$

auspec
spectral dynamics (FFT)

f_{auspec}

MLP $d \rightarrow 96 \rightarrow 48 \rightarrow 1$ · CCC loss · mixup · 50 seeds (avg)

4a Weighted Fusion



$$\hat{y}_{\text{raw}} = 0.40 f_{\text{orig}} + 0.30 f_{\text{auvel}} + 0.20 f_{\text{e3only}} + 0.10 f_{\text{fatigue}} + 0.05 f_{\text{auspec}}$$
 $\hat{y}_{\text{raw}}$

raw (test)

moment match

train PHQ-9

$$\hat{y} = \text{clip}((\hat{y}_{\text{raw}} - \mu) / \sigma \cdot \sigma_t + \mu_t, 0, 27)$$

$\mu = 4.82$ $\sigma = 3.92$

4c Label Rule (confirmation)

The diagram shows two input boxes on the left. The top box contains the expression $\hat{y} \geq t_1 (4.25)$. The bottom box contains the expression $f_{orig} \geq 4$ (support). Arrows from both boxes point to a central blue rounded square labeled "AND". An arrow points from the "AND" box to the right, leading to the text "Binary (0 / 1)".

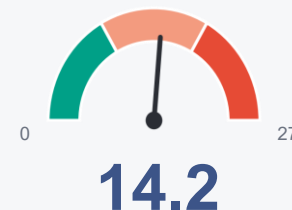
ternary thresholds



0 4.25 11 27

ternary forced to 0 when binary = 0

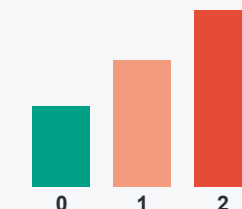
PHQ-9
continuous 0–27



Binary (0 / 1)



Ternary (0 / 1 / 2)



Raw modalities
high-dimensional

- Feature cutting
 - pruning & semantic selection

Predictors
lightweight rankers · seed averaging

- Fusion & calibration**
 - metric-aware · confirmation rule

- Outputs**
 - continuous score & discrete labels